SOUNDNESS OF THE $K = 8$ ROTATIONAL ENCODINGS: DRAFT SECTION

1. Soundness of the encodings for the $k = 8$ rotational upper bounds

The sixfold and sevenfold upper bounds for $k = 8$ rest on unsatisfiability certificates for CNF formulas written by two programs [1]:

- *encoder B*, `C6UPPER/encb/gen_b2.py` (SHA-256 `3d47459a...`), which decides the angular configuration of every triple by integer comparisons; at $(m, K, t) = (7, 8, 10)$ it reproduces byte for byte the certified encoder-B formula of `C7UPPER` (SHA-256 `b4d29017...`, written by `C7UPPER/gen_b.py`; exact computation, `C6UPPER` §3b);
- *encoder A*, `C7UPPER/gen.py` (SHA-256 `4564e7f0...`), which classifies one representative phase vector per pattern in binary64 arithmetic.

This section proves that the formulas are *sound*: every real configuration that the upper bounds exclude yields a satisfying assignment. Encoder B is the primary object; encoder A is treated as a cross-check (Proposition 1.25). The instances are $(m, K, t) = (6, 8, 11)$ and $(7, 8, 10)$; the lemmas are stated for $m \geq 3$ and $4 \leq K \leq 2m$ where this costs nothing, except that the soundness of encoder A is proved only for $m \in \{6, 7\}$. The certificates themselves, and which of their cake_lpr verdicts have been replayed against hashed files, are described in the section on rotational upper bounds for $k = 8$ (draft `STATUS20261104/k8_section.tex`); here we prove only the implication from unsatisfiability to geometry.

Rigor labels. *Lean-checked* items are in `LAYERLEAN/LayerLemma.lean` (SHA-256 `afb8542d...`, the file as committed in `393619a57`; namespace `ErdosSzekeres.LayerLemma`; Lean 4.24.0, Mathlib `f897ebcf`; the axiom check `LAYERLEAN/logs/axioms.log` reports only `propext`, `Classical.choice`, `Quot.sound` for every cited theorem of that file); the declaration names below were checked against that file, and Lean was not rerun for this section. *Written proof (not refereed)* means a complete proof given here. Most of them expand arguments reviewed on 4 November 2026 with no blocking gap found (`C6C7REVIEW/SOUNDNESS.md`). The others have had no internal review: those marked *new here* are new arguments, and those marked *not in the review* are standard facts or assembly steps that the review does not contain. Line numbers refer to `gen_b2.py` with the SHA-256 above. Statements about what a program emits are established by reading its source (Section 1.10, gap (G1)).

1.1. **Configurations and layers.** Let O be the origin, $e(\varphi) = (\cos\varphi, \sin\varphi)$, and ρ the counterclockwise rotation by $2\pi/m$ about O (the Lean `rotAbout` with centre O and angle $2\pi/m$). As in the rest of the note, $\operatorname{orient}(p, q, r) = (x_q - x_p)(y_r - y_p) - (y_q - y_p)(x_r - x_p)$ (the Lean `orient`). It is invariant under translations and rotations, changes sign under reflections, and $\operatorname{orient}(p_{\pi(1)}, p_{\pi(2)}, p_{\pi(3)}) = \operatorname{sgn}(\pi)\operatorname{orient}(p_1, p_2, p_3)$ for every permutation π. A finite set X is in *general position* if no three of its points are collinear (Lean `GeneralPosition`). It is in *convex position* if it is convex independent: Lean `ConvexPosition` is defined as Mathlib's `ConvexIndependent` of the inclusion of X, and by Mathlib's `convexIndependent_set_iff_notMem_convexHull_diff` this holds if and only if $x \notin \operatorname{conv}(X \setminus \{x\})$ for every $x \in X$. A set S is *K-free* if no K-element subset of S is in convex position, that is, if $\neg$ `HasConvexNGon` K S. For a finite set X let $\operatorname{hv}(X) = \{x \in X : x \notin \operatorname{conv}(X \setminus \{x\})\}$ (Lean `hullVerts`), and let $\operatorname{peel}(X, 0) = X$, $\operatorname{peel}(X, n+1) = \operatorname{peel}(X, n) \setminus \operatorname{hv}(\operatorname{peel}(X, n))$ (Lean `peel`).

Definition 1.1 (nested regular m-gons)**.** A *set of t nested regular m-gons* is a set $S = L_0 \cup \cdots \cup L_{t-1}$, where each $L_i = \{\rho^k x_i : k \in \mathbb{Z}\}$ is the orbit of a point $x_i \neq O$ (a regular m-gon centred at O with m vertices), the L_i are pairwise disjoint, and $L_j \subseteq \operatorname{conv}(L_i)$ for $i < j$. Write $r_i = |x_i|$. General position and K-freeness are separate hypotheses.

Lemma 1.2 (layer lemma; Lean-checked (`layers_are_orbits`, `card_peel`, `hullVerts_eq_orbitFS_rotAbout`); the assembly below is a written step, not in the review)**.** *Let $m \geq 1$, $K \leq 2m$ and $t \geq 1$, and let S be a finite ρ-invariant K-free set with $O \notin S$ and $|S| = mt$. Put $L_n = \operatorname{hv}(\operatorname{peel}(S, n))$ for $n < t$. Then each L_n is one ρ-orbit of exactly m points, the L_n are pairwise disjoint, $\operatorname{peel}(S, n) = L_n \cup \cdots \cup L_{t-1}$ (in particular $S = L_0 \cup \cdots \cup L_{t-1}$), and $L_j \subseteq \operatorname{conv}(L_i)$ for $i < j$. So S, with these layers, is a set of t nested regular m-gons.*

Date: Standalone compile check, 6 November 2026.

Proof. For every n with $\mathrm{peel}(S,n) \neq \emptyset$, `layers_are_orbits` gives a point $x \in \mathrm{peel}(S,n)$ (so $x \neq O$) with $\mathrm{hv}(\mathrm{peel}(S,n)) = \{\rho^k x : 0 \le k < m\}$, a set of exactly m points, and $\mathrm{peel}(S,n+1) \subseteq \mathrm{conv}\,\mathrm{hv}(\mathrm{peel}(S,n))$; as $\rho^m = \mathrm{id}$, this set is the orbit $\{\rho^k x : k \in \mathbb{Z}\}$. Since $m(t-1) < |S|$, `card_peel` gives $|\mathrm{peel}(S,n)| = |S| - mn$ for $n \le t$; so $\mathrm{peel}(S,n) \neq \emptyset$ for $n < t$ and $\mathrm{peel}(S,t) = \emptyset$. By definition, $\mathrm{peel}(S,n)$ is the disjoint union of L_n and $\mathrm{peel}(S,n+1)$, since $\mathrm{hv}(X) \subseteq X$. Downward induction from $\mathrm{peel}(S,t) = \emptyset$ gives $\mathrm{peel}(S,n) = L_n \sqcup \cdots \sqcup L_{t-1}$, a disjoint union. For $i < j$, $L_j \subseteq \mathrm{peel}(S,j) \subseteq \mathrm{peel}(S,i+1) \subseteq \mathrm{conv}(L_i)$. □

Definition 1.3 (admissible sets)**.** Let $m \ge 3$, $4 \le K \le 2m$ and $t \ge 1$. A finite set $S \subseteq \mathbb{R}^2$ is (m,K,t)-*admissible* if it is ρ-invariant, $O \notin S$, $|S| = mt$, and S is in general position and K-free. Its *layers* are $L_n = \mathrm{hv}(\mathrm{peel}(S,n))$, $0 \le n < t$. By Lemma 1.2, S with these layers is a set of t nested regular m-gons, so Definition 1.8 below applies to it.

From here on, every configuration is an admissible set with these layers. The Lean theorems are stated for $\mathrm{hv}(\mathrm{peel}(S,n))$ and $\mathrm{peel}(S,n)$, so they apply to the layers with no translation step.

Lemma 1.4 (affine images; written proof (not refereed), routine, not in the review)**.** *Let g be an affine bijection of $\mathbb{R}^2$. Then $\mathrm{conv}(gY) = g\,\mathrm{conv}(Y)$ for every $Y \subseteq \mathbb{R}^2$, so $\mathrm{hv}(gX) = g\,\mathrm{hv}(X)$ and $\mathrm{peel}(gX,n) = g\,\mathrm{peel}(X,n)$ for every finite X and every n; and g preserves general position, convex position and K-freeness. If moreover $g(O) = O$ and $g\rho g^{-1} \in \{\rho,\rho^{-1}\}$, then g maps every (m,K,t)-admissible set S to an (m,K,t)-admissible set whose layers are $g(L_0),\dots,g(L_{t-1})$, in this order.*

Proof. g and g^{-1} map convex combinations to convex combinations, so $g\,\mathrm{conv}(Y) = \mathrm{conv}(gY)$. Hence, for $x \in X$, $x \in \mathrm{conv}(X \setminus \{x\})$ if and only if $gx \in \mathrm{conv}(gX \setminus \{gx\})$; this gives $\mathrm{hv}(gX) = g\,\mathrm{hv}(X)$, and, since the injective map g commutes with set differences, $\mathrm{peel}(gX,n) = g\,\mathrm{peel}(X,n)$ by induction on n. The same equivalence, applied to subsets, shows that g preserves convex position (in the form $x \notin \mathrm{conv}(X \setminus \{x\})$ for all x), hence K-freeness; g and g^{-1} preserve collinearity, hence general position. If $g\rho g^{-1} = \rho^{\pm 1}$, then also $g^{-1}\rho g = \rho^{\pm 1}$, so $\rho(gS) = g\rho^{\pm 1}(S) = gS$; here $\rho^{-1}S = S$ because ρ maps the finite set S injectively into itself. Also $O = g(O) \notin gS$ and $|gS| = |S|$. The layers of gS are $\mathrm{hv}(\mathrm{peel}(gS,n)) = g\,\mathrm{hv}(\mathrm{peel}(S,n)) = g(L_n)$. □

Remark 1.5 (nested m-gons are admissible; written proof (not refereed), new here). Let $m \ge 3$ and $4 \le K \le 2m$. If S is a set of t nested regular m-gons $L_0,\dots,L_{t-1}$ (Definition 1.1) that is in general position and K-free, then S is (m,K,t)-admissible and $\mathrm{hv}(\mathrm{peel}(S,n)) = L_n$ for $n < t$: its layers in the sense of Definition 1.3 are the given ones, in the given order. This is used only to restate Theorem 1.24 for sets presented as nested m-gons. It is not used for Corollary 1.28, so it is not part of the trust base of the upper bounds.

Proof. A nontrivial rotation about O fixes only O, and $x_i \neq O$, so each orbit L_i has exactly m points and avoids O; the L_i are ρ-orbits and pairwise disjoint, so S is ρ-invariant, $O \notin S$ and $|S| = mt$. Let $U_n = L_n \cup \cdots \cup L_{t-1}$; we show $\mathrm{hv}(U_n) = L_n$. Then $\mathrm{peel}(S,n) = U_n$ follows by induction on n, because $U_n \setminus L_n = U_{n+1}$. If $y \in L_j$ with $j > n$, then $y \in \mathrm{conv}(L_n) \subseteq \mathrm{conv}(U_n \setminus \{y\})$, so $y \notin \mathrm{hv}(U_n)$. Let $x \in L_n$ and $f(y) = \langle y, x\rangle$. For $y \in L_n \setminus \{x\}$ we have $f(y) = r_n^2 \cos\gamma < r_n^2 = f(x)$, where $\gamma \not\equiv 0 \pmod{2\pi}$ is the angle between x and y. Hence on $\mathrm{conv}(L_n)$ the value $f(x)$ is attained only at x (a convex combination with value $f(x)$ puts all its weight on x). Every $y \in U_n \setminus \{x\}$ lies in $\mathrm{conv}(L_n)$ and differs from x, so $f(y) < f(x)$; the maximum of the linear function f on the polytope $\mathrm{conv}(U_n \setminus \{x\})$ is attained at a point of $U_n \setminus \{x\}$, hence is $< f(x)$, and $x \notin \mathrm{conv}(U_n \setminus \{x\})$. □

By Lemma 1.2 and Remark 1.5, the (m,K,t)-admissible sets are exactly the K-free sets of t nested regular m-gons in general position, with the same layers. Only the first direction (Lemma 1.2) enters the proofs below.

Lemma 1.6 (centre exclusion; Lean-checked (`centre_not_mem_of_even`))**.** *If m is even and S is ρ-invariant, in general position, and contains a point $p \neq O$, then $O \notin S$.*

Proof. $\rho^{m/2}$ is the half-turn, so $-p \in S$, and $p, O, -p$ are three distinct collinear points. □

Hence a sixfold symmetric set in general position other than $\{O\}$ has $6t$ points, and 61 points force $t \ge 11$; a sevenfold one has $7t$ or $7t+1$ points, and 65 points force $t \ge 10$. The reduction of the two upper bounds to the propositions `Sixfold66` and `Sevenfold70` (every centre-free set of 66, resp. 70, points in general position with sixfold, resp. sevenfold, rotational symmetry has a convex octagon) is Lean-checked: `convexNGon_of_layers`, `sixfold_octagon`, `sevenfold_octagon`. In the language of Definition 1.3, and after translating the centre to O, `Sixfold66` says that no $(6,8,11)$-admissible set exists, and `Sevenfold70` that no $(7,8,10)$-admissible set exists.

Lemma 1.7 (strict nesting and radius order; Lean-checked (`strict_nesting`, `dist2_le_of_later`))**.** *Let S be (m,K,t)-admissible with layers L_n. For $i < j$, $x \in L_i$ and $y \in L_j$ we have $\mathrm{orient}(x,\rho x,y) > 0$; for $i \le j$ we have $r_j \le r_i$.*

Proof. $x \in \mathrm{hv}(\mathrm{peel}(S, i))$ by the definition of L_i, and $y \in L_j \subseteq \mathrm{peel}(S, j)$ (Lean `hullVerts_subset`). Together with ρ-invariance, $O \notin S$, general position, K-freeness, $m \geq 2$ and $K \leq 2m$, these are the hypotheses of `strict_nesting` (layers $n = i < n' = j$) and of `dist2_le_of_later` ($n = i \leq n' = j$, which gives $|y|^2 \leq |x|^2$), both with centre O. □

1.2. **Phases, genericity and the intended assignment.**

Definition 1.8 (phases, labels, genericity)**.** Let S be a set of t nested regular m-gons. The directions of the points of L_i form a coset of $(2\pi/m)\mathbb{Z}$ modulo 2π, which meets $[0, 2\pi/m)$ in exactly one value θ_i, the *phase* of L_i. For $(i, k) \in \Lambda = \{0, \ldots, t-1\} \times \mathbb{Z}/m$ put $P(i, k) = r_i\, e(\theta_i + 2\pi k/m)$; then $(i, k) \mapsto P(i, k)$ is a bijection $\Lambda \to S$ and $\rho P(i, k) = P(i, k+1)$. We order Λ lexicographically, with $k \in \{0, \ldots, m-1\}$. The configuration is *normalised* if $\theta_0 = 0$. Let $u = \pi/m$ for odd m and $u = 2\pi/m$ for even m, and $\psi_i = \theta_i \bmod u \in [0, u)$; for odd m let $h_i \in \{0, 1\}$ with $\theta_i = \psi_i + h_i\pi/m$. The configuration is *generic* if $\theta_i - \theta_j \notin u\mathbb{Z}$ for all $i \neq j$. For a normalised configuration this says that $\psi_1, \ldots, \psi_{t-1}$ are pairwise distinct and nonzero ($\psi_0 = 0$); then $\theta_i \in (0, 2\pi/m)$ for $i \geq 1$, and $\theta_i \neq \pi/m$ when m is odd.

Genericity does not depend on the order of the layers, and it is preserved by rotations about O.

Lemma 1.9 (collinearity with the centre; written proof (not refereed), new here)**.** *A set of t nested regular m-gons is generic if and only if any two distinct points $p, q \in S$ collinear with O satisfy: m is even and $q = -p$ (so p, q lie in one layer and $q = \rho^{m/2}p$).*

Proof. $P(i, k)$ and $P(j, l)$ are collinear with O if and only if their directions differ by a multiple of π, that is, $\theta_i - \theta_j + 2\pi(k-l)/m \in \pi\mathbb{Z}$. Now $\pi\mathbb{Z} + (2\pi/m)\mathbb{Z} = u\mathbb{Z}$: for odd m because $\pi = m \cdot \frac{\pi}{m}$, $\frac{2\pi}{m} = 2 \cdot \frac{\pi}{m}$ and $\gcd(m, 2) = 1$; for even m because $\pi = \frac{m}{2} \cdot \frac{2\pi}{m}$. So for $i \neq j$ some $P(i, k)$, $P(j, l)$ are collinear with O if and only if $\theta_i - \theta_j \in u\mathbb{Z}$. For $i = j$ the condition is $2(k-l) \in m\mathbb{Z}$: for odd m this forces $k = l$, and for even m it gives $k = l$ or $l = k + m/2$, i.e. $P(i, l) = -P(i, k)$. □

Lemma 1.10 (convex position is decided by orientations; written proof (not refereed), standard, not in the review)**.** *Let X be finite and in general position, and $x \in X$. Then $x \in \mathrm{conv}(X \setminus \{x\})$ if and only if there are distinct $a, b, c \in X \setminus \{x\}$ such that* $\mathrm{orient}(b, c, x)$, $\mathrm{orient}(c, a, x)$ *and* $\mathrm{orient}(a, b, x)$ *all have the sign of* $\mathrm{orient}(a, b, c)$. *Consequently, if a bijection between two finite sets in general position preserves the sign of* orient *on all triples, it maps subsets in convex position to subsets in convex position.*

Proof. By Carathéodory's theorem in the plane, $x \in \mathrm{conv}(Y)$ implies $x \in \mathrm{conv}(T)$ for some $T \subseteq Y$ with $|T| \leq 3$. If $|T| \leq 2$, then x equals a point of T (impossible, $x \notin Y$) or lies on a segment between two points of X, contradicting general position. So $x \in \mathrm{conv}\{a, b, c\}$ with $\mathrm{orient}(a, b, c) \neq 0$. Write $x = \lambda_a a + \lambda_b b + \lambda_c c$ with $\lambda_a + \lambda_b + \lambda_c = 1$. Since $\mathrm{orient}(b, c, \cdot)$ is affine and vanishes at b, c, $\mathrm{orient}(b, c, x) = \lambda_a\, \mathrm{orient}(b, c, a) = \lambda_a\, \mathrm{orient}(a, b, c)$; likewise $\mathrm{orient}(c, a, x) = \lambda_b\, \mathrm{orient}(a, b, c)$ and $\mathrm{orient}(a, b, x) = \lambda_c\, \mathrm{orient}(a, b, c)$. Thus $x \in \mathrm{conv}\{a, b, c\}$ iff all $\lambda \geq 0$, and by general position no λ vanishes; this is the stated sign condition. Conversely, if the sign condition holds for distinct $a, b, c \in X \setminus \{x\}$, then $\mathrm{orient}(a, b, c) \neq 0$, so x has barycentric coordinates $\lambda_a, \lambda_b, \lambda_c$ with respect to a, b, c; the same identities show that they are positive, so $x \in \mathrm{conv}\{a, b, c\} \subseteq \mathrm{conv}(X \setminus \{x\})$. For the consequence, let $f : X \to X'$ be the bijection and $Y \subseteq X$. Subsets of a set in general position are in general position, so by the first part and the characterisation of convex position recalled in Section 1.1, Y is in convex position if and only if no $y \in Y$ and distinct $a, b, c \in Y \setminus \{y\}$ satisfy the sign condition. That condition refers only to signs of orient on triples of Y, which f preserves; so Y is in convex position if and only if $f(Y)$ is. □

Lemma 1.11 (genericity; written proof (not refereed))**.** *Let $m \geq 3$, $4 \leq K \leq 2m$.*

- *(a) If m is even, every set of t nested regular m-gons in general position is generic.*
- *(b) If m is odd and there is an (m, K, t)-admissible set, then there is a generic, normalised (m, K, t)-admissible set (with its layers as in Definition 1.3).*

Proof. (a) If $\theta_i - \theta_j \in u\mathbb{Z}$ with $i \neq j$, the proof of Lemma 1.9 gives k, l with $P(i, k)$, $P(j, l)$ collinear with O; then $P(i, k)$, $P(j, l)$ and $P(j, l + m/2) = -P(j, l)$ are three distinct points of S on one line (the first is not in L_j), contradicting general position. This is the mechanism of Lemma 1.6; no perturbation is needed.

(b) Let S be (m, K, t)-admissible, with layers L_i, radii r_i and phases θ_i (Lemma 1.2, Definition 1.8). For $\delta = (\delta_1, \ldots, \delta_{t-1}) \in \mathbb{R}^{t-1}$ and $\delta_0 = 0$ put $P_\delta(i, k) = r_i\, e(\theta_i + \delta_i + 2\pi k/m)$ and $S_\delta = \{P_\delta(\lambda) : \lambda \in \Lambda\}$. For distinct labels λ, μ, ν the function $\delta \mapsto \mathrm{orient}(P_\delta(\lambda), P_\delta(\mu), P_\delta(\nu))$ is continuous and is nonzero at $\delta = 0$, by general position of S. There are finitely many label triples, so there is $\varepsilon > 0$ such that for $\|\delta\|_\infty < \varepsilon$ all these orientations have their signs at $\delta = 0$. Fix such a δ. Then: (1) distinct labels give distinct points, since $P_\delta(\lambda) = P_\delta(\mu)$ would

make $\mathrm{orient}(P_\delta(\lambda), P_\delta(\mu), P_\delta(\nu)) = 0$ for any third label ν; so $|S_\delta| = mt$; (2) S_δ is in general position; (3) S_δ is ρ-invariant and $O \notin S_\delta$ (each L_i is rotated as a whole and $r_i > 0$); (4) by Lemma 1.10 applied to $P(\lambda) \mapsto P_\delta(\lambda)$ and its inverse, S_δ is K-free. So S_δ is (m, K, t)-admissible, and by Lemma 1.2 its layers are ρ-orbits. Its ρ-orbits are the rotated layers $\{P_\delta(i,k) : k\}$ (the orbits of ρ on a set are determined by the set), so its layers are these orbits in some order (in fact in the original order, because the nesting signs of Lemma 1.7 are preserved, but this is not needed), and their phases are $\theta_i + \delta_i \bmod 2\pi/m$. Since genericity does not depend on the order of the layers, S_δ is generic as soon as

$$\delta \notin B = \bigcup_{0 \le i < j \le t-1} \{\delta : \ \delta_i - \delta_j \in \theta_j - \theta_i + u\mathbb{Z}\} \qquad (\delta_0 = 0).$$

For each pair $i < j$, the values $\delta_i - \delta_j$ on the box $(-\varepsilon, \varepsilon)^{t-1}$ lie in an interval of length 4ε, which meets $\theta_j - \theta_i + u\mathbb{Z}$ in finitely many points c; each set $\{\delta_i - \delta_j = c\}$ (or $\{\delta_j = -c\}$ when $i = 0$) is a proper affine hyperplane of $\mathbb{R}^{t-1}$. A finite union of proper hyperplanes has empty interior, so the box contains a $\delta \notin B$. Finally rotate S_δ about O so that its outermost layer has phase 0: by Lemma 1.4 (a rotation about O commutes with ρ) the result is admissible with the rotated layers in the same order, and since all phases change by the same constant modulo $2\pi/m$, it is still generic. (If $t = 1$ there is nothing to perturb.) □

For odd m the perturbation is needed even though S is in general position: a point and another point on the opposite ray from O are collinear with O, which is not a point of S (the centre of an original sevenfold set is removed in the reduction). By Lemma 1.9, after the perturbation no two points of S are collinear with O.

Definition 1.12 (intended assignment)**.** Let S be a generic, normalised (m, K, t)-admissible set, with its layers $L_n = \mathrm{hv}(\mathrm{peel}(S, n))$. Its *intended assignment* α_S gives

- the phase-order variable $Y(i,j)$, $1 \le i < j \le t-1$, the value $[\psi_i < \psi_j]$;
- for odd m, the variable $H(i)$, $1 \le i \le t-1$, the value $[\theta_i < \pi/m] = [h_i = 0]$;
- the orientation variable v_c of a triple orbit with representative $c = (c_0, c_1, c_2)$ (sorted in Λ; Definition 1.13), the value $[\mathrm{orient}(P(c_0), P(c_1), P(c_2)) > 0]$;
- the chain variables the values defined in Section 1.5 (before Lemma 1.21).

In both encoders $\psi_0 = 0$ is the least phase and $h_0 = 0$. Encoder B gives layer 0 rank 0 and $h = 0$ (lines 91 and 94). In encoder A, `Enc.patterns` keeps only the orders that put layer 0 first, gives it the phase `0.0` and no H literal (`gen.py` lines 190, 211 and 218–219), and `ylit` returns the constants `True`/`False` for pairs involving layer 0 (lines 125–135); the helper `hlit` is never called. Encoder B numbers the variables $Y(i,j)$ first (lines 36–39, lexicographically in (i,j)), then $H(i)$ for odd m (line 40), then the orientation variables (lines 50–55), then the chain variables on demand (lines 161–165). Encoder A numbers the orientation variables first, then Y, then H, then the chain variables (`Enc.__init__`, `fam_chain`).

1.3. **Triple orbits and orientation variables.**

Definition 1.13 (triple orbits, orientation variables)**.** The rotation acts on Λ by $(i,k) \mapsto (i, k+1)$ and on its 3-element subsets; the orbits of the latter are the *triple orbits*. Each encoder has one *orientation variable* v_c per triple orbit, indexed by a representative $c = (c_0, c_1, c_2)$ sorted in Λ, meant to express $\mathrm{orient}(P(c_0), P(c_1), P(c_2)) > 0$.

Encoder B's representative `canon`(T) (lines 42–49) is the lexicographically least of the sorted lists $\rho^{-k}T$, $(i,k) \in T$; one variable is allocated per value of `canon` (lines 50–55). For an ordered triple of distinct labels, `olit` (lines 63–76) computes $c =$`canon`$(\{a,b,d\})$, takes the first point (i_0, k_0) of the list $[a,b,d]$ such that the sorted list of $\rho^{-k_0}[a,b,d]$ equals c, puts $rt = \rho^{-k_0}[a,b,d]$ and lets ω be the permutation with $c_j = rt_{\omega(j)}$; it returns v_c if $\operatorname{sgn}\omega = +1$ and $\neg v_c$ otherwise.

Lemma 1.14 (one variable per triple orbit; written proof (not refereed))**.** *(a)* `canon` *is constant on each triple orbit, and* `canon`(T) *lies in the orbit of* T*; so encoder B's orientation variables correspond bijectively to the triple orbits.*

(b) Under α_S*, the literal* `olit`(a,b,d) *is true if and only if* $\mathrm{orient}(P(a), P(b), P(d)) > 0$*, for every ordered triple of distinct labels.*

(c) If ρ^s *with* $s \not\equiv 0 \pmod m$ *maps a 3-set* T *onto itself, then* $3 \mid m$*,* $T = \{x, \rho^{m/3}x, \rho^{2m/3}x\}$ *lies in one layer, and* ρ^s *permutes* T *cyclically. Consequently the sign computed in (b) does not depend on which admissible rotation is used.*

For $(m,t) = (6,11)$ *there are* 45,760 *triples: the* 22 *triples* $\{(i,k),(i,k+2),(i,k+4)\}$ *form* 11 *orbits of size* 2 *and the others* 7,623 *orbits of size* 6*, in total* 7,634 *orbits. For* $(7,10)$ *all* 7,820 *orbits of the* 54,740 *triples*

have size 7 *(exact computation,* `k8_soundness_checks.py`*; the totals agree with the review's independent count,* `C6C7REVIEW/SOUNDNESS.md`*).*

Proof. (a) The candidates for $\rho^r T$ are the lists $\rho^{-(k+r)}\rho^r T = \rho^{-k}T$, $(i,k) \in T$, i.e. the candidates for T. So $\texttt{canon}(\rho^r T) = \texttt{canon}(T)$, and $\texttt{canon}(T)$ is the sorted list of some $\rho^{-k}T$. Triples in different orbits therefore have different representatives.

(b) The loop of `olit` succeeds because `canon` is attained by one of the candidates. Let $s = -k_0$. Rotations about O preserve orient and $P(\rho^s\lambda) = \rho^s P(\lambda)$, so

$$\operatorname{orient}(P(a), P(b), P(d)) = \operatorname{orient}(P(rt_0), P(rt_1), P(rt_2)) = \operatorname{sgn}(\omega)\, \operatorname{orient}(P(c_0), P(c_1), P(c_2)),$$

because (c_0, c_1, c_2) is the permutation ω of (rt_0, rt_1, rt_2) and a permutation and its inverse have the same sign. All orientations are nonzero by general position, so $\texttt{olit}(a, b, d)$, which is v_c or $\neg v_c$ according to $\operatorname{sgn}\omega$, is true under α_S exactly when $\operatorname{orient}(P(a), P(b), P(d)) > 0$. The argument holds for every admissible choice of s.

(c) A rotation $\rho^s \neq \mathrm{id}$ fixes no point of Λ (it fixes only O), so it permutes T without fixed points, i.e. as a 3-cycle. Then ρ^{3s} fixes T pointwise, so $\rho^{3s} = \mathrm{id}$ and $m \mid 3s$ while $m \nmid s$; hence $3 \mid m$ and $s \equiv \pm m/3$. Rotations preserve layers, so $T \subseteq L_i$ for one i and $T = \{x, \rho^{m/3}x, \rho^{2m/3}x\}$. Two admissible rotations s, s' for T differ by a rotation fixing the set $\rho^s T = c$; the two lists rt then differ by the identity or a 3-cycle, both even, so $\operatorname{sgn}\omega$ is the same. For $m = 6$ these are the triangles $\{(i,k), (i,k+2), (i,k+4)\}$, $k \in \{0,1\}$, two per layer, each orbit of size $6/3 = 2$; for $m = 7$ there are none. The counts follow: $(\binom{66}{3} - 22)/6 = 7{,}623$ and $\binom{70}{3}/7 = 7{,}820$. □

Part (c) is not needed for (b); it shows that the literal map never identifies an orientation variable with its own negation, which is how the stabiliser question was raised in review. Encoder A's representative is the least sorted list $\rho^s T$ over all $s \in \mathbb{Z}/m$, and its literal function `Enc.o` is correct by the same computation as (b). Its representatives coincide with encoder B's (exact computation over all 45,760 and 54,740 triples, 0 mismatches; `C6C7REVIEW/SOUNDNESS.md`); soundness does not use this.

1.4. **Directions and the angular classification.** For $v \neq 0$ let $\operatorname{dir}(v) \in [0, 2\pi)$ be its direction. Let M be a set of layer indices, each $i \in M$ with a phase $\theta_i \in [0, 2\pi/m)$, such that the values $\psi_i = \theta_i \bmod u$, $i \in M$, are pairwise distinct; for odd m define h_i by $\theta_i = \psi_i + h_i\pi/m$ as in Definition 1.8. Only directions matter below, so when no configuration is given we put $P(i,k) = e(\theta_i + 2\pi k/m)$. The two cases used are the layers of a generic configuration, with its points $P(i,k)$ (normalised or not), and, in Proposition 1.25, the layers of one triple with the phases sampled by encoder A. The *objects* of a label (i,k), $i \in M$, are the point $P(i,k)$ and its antipode $-P(i,k)$ (a direction, not necessarily a point of S). Put $N = 2m$ for odd m and $N = m$ for even m, and define the cell index $J \in \{0, \dots, N-1\}$ by

$$J(P(i,k)) = (2k + h_i) \bmod 2m, \quad J(-P(i,k)) = (J(P(i,k)) + m) \bmod 2m \qquad (m \text{ odd}),$$

$$J(P(i,k)) = k, \quad J(-P(i,k)) = (k + m/2) \bmod m \qquad (m \text{ even}).$$

A *rank* is a map $\mathrm{rk} : M \to \mathbb{Z}$ with $\mathrm{rk}(i) < \mathrm{rk}(j) \iff \psi_i < \psi_j$; the *key* of an object ω of layer $i \in M$ is $\kappa(\omega) = (J(\omega), \mathrm{rk}(i))$, ordered lexicographically. These are the keys of lines 104–111.

Lemma 1.15 (key order; written proof (not refereed))**.** *Let M, the phases θ_i ($i \in M$) and a rank be as above. For objects ω of layer $i \in M$, $\operatorname{dir}(\omega) = uJ(\omega) + \psi_i$. Hence $\operatorname{dir}(\omega) < \operatorname{dir}(\omega')$ if and only if $\kappa(\omega) < \kappa(\omega')$, and $\operatorname{dir}(\omega) = \operatorname{dir}(\omega')$ if and only if $\kappa(\omega) = \kappa(\omega')$. Two distinct objects have equal keys if and only if m is even and they are $P(i,k)$ and $-P(i,k+m/2)$.*

Proof. For odd m, $\theta_i + 2\pi k/m = \psi_i + (\pi/m)(2k + h_i)$, and reducing modulo $2\pi = 2m \cdot \pi/m$ leaves $\psi_i + (\pi/m)J$ with $J = (2k + h_i) \bmod 2m$, a value in $[0, 2\pi)$ since $\psi_i < \pi/m$; the antipode adds $\pi = m \cdot \pi/m$. For even m, $\theta_i + 2\pi k/m = \psi_i + (2\pi/m)k$ with $0 \le k < m$, and the antipode adds $\pi = (m/2) \cdot 2\pi/m$. The map $(J, \psi) \mapsto uJ + \psi$ from $\{0, \dots, N-1\} \times [0, u)$ to $[0, 2\pi)$ is strictly increasing for the lexicographic order (if $J < J'$ then $uJ + \psi < u(J+1) \le uJ' + \psi'$), and replacing ψ by a rank does not change the lexicographic order. Equal keys mean equal J and equal rank, i.e. (the ψ_i, $i \in M$, are distinct) the same layer. Within one layer, two points or two antipodes with equal J have equal k, and a point and its own antipode have J differing by m or $m/2$. A point $P(i,k)$ and an antipode $-P(i,l)$ with equal J satisfy, for odd m, $2(k - l) \equiv m \pmod{2m}$, impossible since m is odd; for even m, $k \equiv l + m/2$. □

We call the two orientation rules *rule A* (centre inside) and *rule B* (half-plane), as in the Lean docstrings; the letters are unrelated to the names of the encoders.

Lemma 1.16 (circular dichotomy; written proof (not refereed) for (a), Lean-checked (`orient_polar`, `orient_polar_centre_inside`, `orient_polar_halfplane`, `sin_add_lt`) for the signs in (b) and (c))**.** *Let $p_1, p_2, p_3 \neq O$ be such that the six directions* $\mathrm{dir}(p_a)$ *and* $\mathrm{dir}(p_a) + \pi$ *are pairwise distinct modulo* 2π*. Read the six objects counterclockwise as a cyclic word in the letters* `P` *(point) and* `A` *(antipode).*

(a) The word alternates, or the three letters `P` *are cyclically consecutive (a rotation of* `PPPAAA`*); the case is determined by the word.*

(b) (rule A) If it alternates, then $\mathrm{orient}(p_{\sigma(1)}, p_{\sigma(2)}, p_{\sigma(3)}) > 0$*, where* $p_{\sigma(1)}, p_{\sigma(2)}, p_{\sigma(3)}$ *are the points in counterclockwise order.*

(c) (rule B) If it is a rotation of `PPPAAA` *and a, b, d are the points in counterclockwise order from the first* `P` *of the block, then the counterclockwise angles x from* $\mathrm{dir}\, a$ *to* $\mathrm{dir}\, b$ *and y from* $\mathrm{dir}\, b$ *to* $\mathrm{dir}\, d$ *satisfy $x, y > 0$ and $x + y < \pi$; if moreover $|b| \geq |a|$ and $|b| \geq |d|$, then* $\mathrm{orient}(a, b, d) > 0$*.*

Proof. (a) Let $\alpha_1, \alpha_2, \alpha_3$ be the directions of the points in counterclockwise order, and $g_1, g_2, g_3 > 0$ the counterclockwise gaps (g_a from α_a to α_{a+1}, indices modulo 3), so $g_1 + g_2 + g_3 = 2\pi$. No gap equals π, since then a point and the antipode of another would coincide. *Case 1: all $g_a < \pi$.* Then $g_a < \pi < g_a + g_{a+1}$ (because $g_{a+2} < \pi$), so $\alpha_a + \pi$ lies in the open arc from α_{a+1} to α_{a+2}. Each of the three open arcs between consecutive points thus contains the antipode of the opposite point, and there are only three antipodes: the word alternates. *Case 2: some gap exceeds π*; at most one can, and after a cyclic relabelling it is g_3, so $g_1 + g_2 < \pi$. Every α_a lies on the closed arc of length $g_1 + g_2$ starting at α_1, so every antipode lies on the closed arc of length $g_1 + g_2$ starting at $\alpha_1 + \pi$. That arc lies inside the open arc from $\alpha_3 = \alpha_1 + g_1 + g_2$ to $\alpha_1 + 2\pi$, because $\alpha_1 + g_1 + g_2 < \alpha_1 + \pi$ and $\alpha_1 + \pi + g_1 + g_2 < \alpha_1 + 2\pi$. So the word is `PPPAAA` up to rotation, the block begins at α_1, and $x = g_1$, $y = g_2$. The two cases give different words.

(b) Case 1. Choose representatives $\alpha < \beta < \delta < \alpha + 2\pi$ of the three directions in counterclockwise order. Then $\beta - \alpha$, $\delta - \beta$ and $\alpha - \delta + 2\pi$ are the gaps, all in $(0, \pi)$, and `orient_polar_centre_inside` (with the polar form `orient_polar`) gives orient > 0; cyclic rotation of the arguments does not change orient.

(c) Case 2 with $\alpha = \alpha_1$, $\beta = \alpha + x$, $\delta = \alpha + x + y$: `orient_polar_halfplane` needs $|a| \leq |b|$, $|d| \leq |b|$, $\beta - \alpha > 0$, $\delta - \beta > 0$ and $\delta - \alpha < \pi$; its proof uses $\sin x + \sin y > \sin(x + y)$ (`sin_add_lt`), as

$$\mathrm{orient}(a, b, d) = |a||b| \sin x + |b||d| \sin y - |d||a| \sin(x + y) \;\geq\; |a||d| \big(\sin x + \sin y - \sin(x + y)\big) > 0. \qquad \square$$

The case in which the six directions are not distinct is excluded by hypothesis; for a generic configuration it occurs exactly for a same-layer antipodal pair with even m (Lemma 1.15). The written dichotomy in the docstring of `gen.py` omitted this third case; it is treated in Section 1.8.

Patterns. Let c be an orbit representative with layer set M_c and free layers $F_c = M_c \setminus \{0\}$. A *pattern* $\pi = (\mathrm{rk}, h)$ consists of a bijection $\mathrm{rk} : F_c \to \{1, \ldots, |F_c|\}$, extended by $\mathrm{rk}(0) = 0$, and for odd m a map $h : F_c \to \{0, 1\}$, extended by $h(0) = 0$. Encoder B enumerates all patterns (lines 89–94) and forms the premise list $\mathrm{prem}(\pi)$ (lines 96–102): for $a < b$ in F_c the literal $Y(a, b)$ if $\mathrm{rk}(a) < \mathrm{rk}(b)$ and $\neg Y(a, b)$ otherwise, and for odd m and $L \in F_c$ the literal $H(L)$ if $h(L) = 0$ and $\neg H(L)$ otherwise. The *real pattern* π_S has $\mathrm{rk}_S(L) = 1 + \#\{L' \in F_c : \psi_{L'} < \psi_L\}$ and $h_S(L) = h_L$.

Lemma 1.17 (premises; written proof (not refereed))**.** *All literals of* $\mathrm{prem}(\pi)$ *are true under α_S if and only if $\pi = \pi_S$. Moreover* rk_S *is a rank on M_c in the sense above.*

Proof. The ψ_L, $L \in F_c$, are distinct and positive (genericity), so rk_S is a bijection onto $\{1, \ldots, |F_c|\}$ and is strictly increasing in ψ also with $\mathrm{rk}_S(0) = 0$, $\psi_0 = 0$. Under α_S, the pair literal for $a < b$ is true iff $[\mathrm{rk}(a) < \mathrm{rk}(b)] = [\psi_a < \psi_b]$, and the H literal iff $h(L) = h_L$. All pairwise comparisons determine a bijection onto $\{1, \ldots, |F_c|\}$ uniquely. $\square$

Lemma 1.18 (angular clauses of encoder B; written proof (not refereed))**.** *Every clause emitted at lines 82–141 of* `gen_b2.py` *is satisfied by α_S.*

Proof. Fix an orbit representative c and its variable $v = v_c$. For each pattern π the code computes a value $\ell(\pi)$, either an orientation literal or `None`, and then (lines 134–141) emits the unit clause $[\ell]$ if all patterns give the same literal $\ell \neq$`None`, and otherwise the clause $\neg\mathrm{prem}(\pi) \vee \ell(\pi)$ for every π with $\ell(\pi) \neq$`None`. We show that $\ell(\pi_S)$ is `None` or true under α_S. This suffices: π_S is among the enumerated patterns, so a unit clause equals $[\ell(\pi_S)]$; and a clause for $\pi \neq \pi_S$ contains the negation of a premise literal that is false under α_S (Lemma 1.17).

For $\pi = \pi_S$ the keys of lines 104–111 are the keys κ of the six objects of the labels in c, with the rank rk_S (Lemma 1.17). If two keys coincide (line 113), then m is even and c contains a same-layer antipodal pair

(Lemma 1.15), and $\ell(\pi_S) =$`None`. Otherwise the six directions are distinct, and sorting the keys (line 118) lists the objects by increasing direction from direction 0 (Lemma 1.15); the string `kinds` is the cyclic word of Lemma 1.16.

- If `kinds` alternates (line 121), `seq` $= (\sigma(1), \sigma(2), \sigma(3))$ lists the points counterclockwise, and $\ell = v$ if $\operatorname{sgn}\sigma = +1$, else $\neg v$. By Lemma 1.16(b),
$$\operatorname{orient}(P(c_0), P(c_1), P(c_2)) = \operatorname{sgn}(\sigma)\ \operatorname{orient}(P(c_{\sigma(1)}), P(c_{\sigma(2)}), P(c_{\sigma(3)}))$$
has the sign $\operatorname{sgn}\sigma$, so ℓ is true under α_S.
- Otherwise the word is a rotation of `PPPAAA` (Lemma 1.16(a)). In the doubled word the substring `PPP` occurs exactly where the block starts, so `dbl.find('PPP')` (line 127) is the position of its first `P`, and (a, b, d) (lines 129–130) are the points in counterclockwise order from there. If $\operatorname{layer}(b) \le \operatorname{layer}(a), \operatorname{layer}(d)$ (line 131), Lemma 1.7 gives $r_{\operatorname{layer}(b)} \ge r_{\operatorname{layer}(a)}, r_{\operatorname{layer}(d)}$, Lemma 1.16(c) gives $\operatorname{orient}(P(a), P(b), P(d)) > 0$, and $\ell =$`olit`(a, b, d) is true by Lemma 1.14(b). If the condition fails, $\ell =$`None`.

The assertions at lines 114, 117 and 128 are not used; they held when the formulas were generated. □

Exhaustively over all orbits and patterns, the six integer keys are not distinct in 622 cases (all for $m = 6$), alternate in 7,901 ($m = 6$) and 56,630 ($m = 7$) cases, form a `PPPAAA` block in 23,681 and 167,997 cases, and never do anything else (exact computation, `k8_soundness_checks.py`; a check of Lemma 1.16(a), not part of the proof). For $m = 6$, $t = 11$ encoder B emits 2,347 angular units and 7,740 conditional angular clauses (`C6UPPER` §3b).

1.5. **Phase-order, nesting and chain clauses.**

Lemma 1.19 (phase order; written proof (not refereed))**.** *The clauses $\neg Y(i,j) \vee \neg Y(j,l) \vee Y(i,l)$ and $Y(i,j) \vee Y(j,l) \vee \neg Y(i,l)$, $1 \le i < j < l \le t-1$ (lines 79–81), are satisfied by α_S.*

Proof. The ψ_i, $i \ge 1$, are distinct reals, so $\neg Y(i,j)$ means $\psi_j < \psi_i$. The first clause is transitivity of $<$. If the second failed, then $\psi_j < \psi_i$, $\psi_l < \psi_j$ and $\psi_i < \psi_l$, a cycle. □

These clauses say that the Y variables form a strict total order; they are not used in the proofs of the other families, which only evaluate α_S.

Lemma 1.20 (nesting units; Lean-checked (`strict_nesting`) for the geometry, written proof (not refereed) for the code)**.** *The unit clauses* `olit`$((i,0),(i,1),(j,k))$, $0 \le i < j \le t-1$, $k \in \mathbb{Z}/m$ *(lines 143–150), are satisfied by α_S.*

Proof. $P(i,1) = \rho P(i,0)$ and $P(j,k) \in L_j$, so Lemma 1.7 gives $\operatorname{orient}(P(i,0), P(i,1), P(j,k)) > 0$; apply Lemma 1.14(b). The check at lines 148–149 only confirms that no nesting unit contradicts an angular unit. □

Chains. Fix a layer i, the pivot $p = (i,0)$ and $Q_i = \{(j,k) : j \ge i\} \setminus \{p\}$ (line 159). A *p-chain of length $l \ge 2$* is a sequence $q_1, \dots, q_l$ in Q_i with $\operatorname{orient}(P(p), P(q_s), P(q_{s+1})) > 0$ for $1 \le s < l$ and $\operatorname{orient}(P(q_s), P(q_{s+1}), P(q_{s+2})) > 0$ for $1 \le s \le l-2$. Set $\alpha_S(F_l^{(i)}(a,b)) =$ [there is a p-chain of length l with $q_1 = a$, $q_2 = b$]. With $L = K-1$, lines 166–190 emit, for the pairs (a,b) whose literal `olit`(p,a,b) is not the negation of an emitted unit (`okpair`):

- (base) $F_2(a,b) \vee \neg$`olit`(p,a,b);
- (extension) $\neg F_l(b,d) \vee \neg$`olit`$(p,a,b) \vee \neg$`olit`$(a,b,d) \vee F_{l+1}(a,b)$ for $2 \le l \le K-3$;
- (closing) $\neg F_{K-2}(b,d) \vee \neg$`olit`$(p,a,b) \vee \neg$`olit`$(a,b,d)$;

omitting the extension and closing clauses with $a \in \{b,d\}$ or with a turn literal `olit`(a,b,d) that is the negation of an emitted unit. The closing clauses forbid a p-chain of length $K-1$, which together with p would be a convex K-gon.

Lemma 1.21 (chain clauses; Lean-checked (`chain_clauses_sound`) for the geometry, written proof (not refereed) for the code)**.** *Let $m \ge 3$ and $4 \le K \le 2m$. The chain clauses of lines 154–190 are satisfied by α_S.*

Proof. Base clause: if $\operatorname{orient}(P(p), P(a), P(b)) > 0$, then (a,b) is a p-chain of length 2. Extension clause: if $(b, d, q_3, \dots, q_l)$ is a p-chain and $\operatorname{orient}(P(p), P(a), P(b)) > 0$, $\operatorname{orient}(P(a), P(b), P(d)) > 0$, then $(a, b, d, q_3, \dots, q_l)$ is a p-chain of length $l+1$. (Literals are evaluated by Lemma 1.14(b).) Closing clause: otherwise there is a p-chain $(a, b, d, \dots)$ of length $K-1$, i.e. points $q_0, \dots, q_{K-2}$ of $\operatorname{peel}(S,i) \setminus \{P(p)\}$ with $\operatorname{orient}(P(p), q_s, q_{s+1}) > 0$ and $\operatorname{orient}(q_s, q_{s+1}, q_{s+2}) > 0$. Here $P(p) \in L_i = \operatorname{hv}(\operatorname{peel}(S,i))$ by the definition of the layers; each q_s is $P(\lambda)$ for a label $\lambda \in Q_i$ of some layer $j \ge i$, so $q_s \in L_j \subseteq \operatorname{peel}(S,i)$ (Lemma 1.2), and $q_s \ne P(p)$ because $\lambda \ne p$ and P is a bijection. `chain_clauses_sound` (hypotheses $m \ge 3$, $4 \le K \le 2m$, $O \notin S$, ρ-invariance, general position, K-freeness) says that no such chain exists. Its proof: by `lin_pos_of_peel` all points of $\operatorname{peel}(S,i)$ other than $P(p)$

lie strictly on the inner side of the tangent to the circle of radius r_i at $P(p)$; by `hasConvexNGon_of_fan` a pivot and $n \geq 3$ chain points in an open half-plane through the pivot satisfy orient > 0 for every triple of $(x_0, \ldots, x_n) = (P(p), q_0, \ldots, q_{n-1})$ taken in sequence order, i.e. orient$(x_s, x_{s'}, x_{s''}) > 0$ for all $s < s' < s''$ (`fan_positive`; identities `orient_trans_halfplane`, `fan_step`) and are therefore in convex position (`ErdosSzekeres.Suk.convexPosition_of_positive_triples`, in `ErdosSzekeres/Suk/OrientationClique.lean`, covered by the axiom check of `hasConvexNGon_of_fan`); this gives K points in convex position. Omitted clauses only weaken the formula. □

Remark 1.22 (completeness; written proof (not refereed), not used). Conversely, if S has K points X in convex position, the closing clauses fail under α_S. Let i be the least layer met by X and $P(i,k) \in X$; then $\rho^{-k}X$ is in convex position, contains $P(p)$, and its other points have labels in Q_i. Listing $\rho^{-k}X$ counterclockwise along its convex hull from $P(p)$ gives $P(p), q_1, \ldots, q_{K-1}$, and every three vertices of a convex polygon in general position, taken in counterclockwise order, are positively oriented. So $(q_1, \ldots, q_{K-1})$ is a p-chain of length $K-1$. Thus the chain family expresses K-freeness exactly; soundness needs only Lemma 1.21.

Encoder A's chains. `fam_chain` of `gen.py` builds the chains forwards: $R_l(q, q')$ means a p-chain of length l ending with q, q', and it is assigned that truth value. The base, extension and closing clauses are justified as above by appending instead of prepending. Encoder A also uses the emitted angular and nesting units (recorded in `forced`) in two ways. If the unit $[\ell]$ is emitted, it drops the literal $\neg\ell$ from a chain clause; since α_S satisfies $[\ell]$ (Proposition 1.25, for $m \in \{6, 7\}$), it satisfies the shortened clause exactly when it satisfies the full one. If the unit $[\neg\ell]$ is emitted, it omits every chain clause containing $\neg\ell$, as encoder B does; such a clause is subsumed by the unit, and omitting it is a weakening.

1.6. **The reflection unit.**

Lemma 1.23 (reflection; written proof (not refereed))**.** *Let S be a generic, normalised (m, K, t)-admissible set, and $\tau(x, y) = (x, -y)$. Then τS is again a generic, normalised (m, K, t)-admissible set, its layers are $\tau(L_0), \ldots, \tau(L_{t-1})$ in this order, and its phases are $\theta'_0 = 0$ and, for $i \geq 1$, $\theta'_i = 2\pi/m - \theta_i$, with $\psi'_i = u - \psi_i$ and, for odd m, $h'_i = 1 - h_i$. Hence $\alpha_{\tau S}(Y(i,j)) = \neg\alpha_S(Y(i,j))$ for $1 \leq i < j$, and for odd m $\alpha_{\tau S}(H(i)) = \neg\alpha_S(H(i))$. In particular exactly one of S and τS satisfies the unit $Y(1,2)$ (even m, $t \geq 3$), resp. $H(1)$ (odd m, $t \geq 2$).*

Proof. τ is a linear bijection with $\tau(O) = O$ and $\tau\rho\tau^{-1} = \rho^{-1}$. By Lemma 1.4, τS is (m, K, t)-admissible and its layers are $\tau(L_0), \ldots, \tau(L_{t-1})$, in this order; $\tau(L_i)$ is the ρ-orbit of τx_i, a regular m-gon of radius r_i. Since $\tau(r\, e(\varphi)) = r\, e(-\varphi)$, the directions of $\tau(L_i)$ are $-\theta_i + (2\pi/m)\mathbb{Z}$; the phase is 0 for $i = 0$ and $2\pi/m - \theta_i$ for $i \geq 1$, because $\theta_i \in (0, 2\pi/m)$. Differences of phases change sign modulo $2\pi/m \in u\mathbb{Z}$, so genericity is kept. For even m, $\psi'_i = \theta'_i = u - \psi_i$. For odd m, $\theta_i = \psi_i + h_i\pi/m$ with $\psi_i \in (0, \pi/m)$ gives $\theta'_i = (\pi/m - \psi_i) + (1 - h_i)\pi/m$ with $\pi/m - \psi_i \in (0, \pi/m)$, so $\psi'_i = \pi/m - \psi_i$ and $h'_i = 1 - h_i$. The map $\psi \mapsto u - \psi$ reverses the order of distinct phases. □

The reflection need not be a symmetry of S: it produces another configuration of the same kind. Layer indices must not be permuted, since they encode the radial order used by rule B and by the nesting and chain clauses; the reflection keeps them.

1.7. **The soundness theorem.**

Theorem 1.24 (soundness; written proof (not refereed), resting on the Lean-checked lemmas cited above)**.** *Let $m \geq 3$, $4 \leq K \leq 2m$ and $t \geq 3$. Let F_B and F_A be the CNF formulas written by* `gen_b2.py` *and* `gen.py` *on input (m, K, t), and let the* reflection unit *be the unit clause $Y(1,2)$ for even m and $H(1)$ for odd m, in each encoder's own numbering. Suppose that some (m, K, t)-admissible set exists, that is, a set of mt points in general position, invariant under ρ, not containing O, with no K points in convex position. By Remark 1.5 this holds in particular if some set of t nested regular m-gons in general position has no K points in convex position. Then:*

(a) F_B plus the reflection unit is satisfiable;
(b) if moreover $m \in \{6, 7\}$, then F_A plus the reflection unit is satisfiable.

Part (b) is restricted to $m \in \{6, 7\}$ because encoder A classifies triples in binary64 arithmetic, and the rounding bound used in Proposition 1.25(i) has been computed only for these two values of m (it does not depend on K or t). For other m the same argument needs that bound for the given m; we make no claim about F_A there. Encoder B uses only integer comparisons, so part (a) needs no such bound.

For the certified instances: at $(6, 8, 11)$ both formulas have 78,059 variables and 2,387,083 clauses (SHA-256 `479e31a9...` for B, `542c8a3a...` for A), and $Y(1,2)$ is variable 1 of F_B (lines 36–39) and variable

7,635 = 7,634 + 1 of F_A; at $(7, 8, 10)$ both have 80,080 variables and 2,748,538 clauses (`b4d29017...` for B, `1a7c0e70...` for A), and $H(1)$ is variable $\binom{9}{2} + 1 = 37$ of F_B and 7,820 + 36 + 1 = 7,857 of F_A. The sevenfold certificate for A is for F_A plus $H(1)$ (`fa297b5b...`, made by `C7UPPER/addunits.py`); the one for B is for F_B alone, which the theorem covers a fortiori. Sizes and hashes: `C6UPPER/README.md`, `C7UPPER/README.md`, `C7UPPER/SHA256SUMS.txt`.

Proof. Let S be an (m, K, t)-admissible set. If m is odd, replace it by the generic, normalised admissible set of Lemma 1.11(b). If m is even, S with its layers is a set of nested regular m-gons (Lemma 1.2) in general position, hence generic by Lemma 1.11(a); rotate it about O so that it is normalised, which keeps it admissible with the rotated layers in the same order (Lemma 1.4) and keeps it generic. By Lemma 1.23, replacing S by τS if necessary, α_S satisfies the reflection unit. (a) Give the variables of F_B their values under α_S (Definition 1.12); every variable of F_B is a Y, H, orientation or chain variable. By reading the source, F_B consists of the header (line 192) and exactly the clauses appended to `cls` at lines 80–81, 137, 141, 147, 175 and 190, that is, the families of Lemmas 1.19, 1.18, 1.20 and 1.21; all are satisfied. (b) Let $m \in \{6, 7\}$ and give the variables of F_A their values under α_S. `build` calls `fam_order`, `fam_angle`, `fam_nest` and `fam_chain` and nothing else (the Grassmann–Plücker family of the docstring is not emitted by the program). The first three are satisfied by Lemma 1.19 (identical clauses) and Proposition 1.25, the last by the paragraph on encoder A's chains. □

Proposition 1.25 (encoder A's angular and nesting clauses; written proof (not refereed), new here, and exact computation)**.** *Let $m \in \{6, 7\}$, $4 \le K \le 2m$, $t \ge 3$, and let S be as in Definition 1.12. Every clause emitted by* `fam_angle` *and* `fam_nest` *of* `gen.py` *on input (m, K, t) is satisfied by α_S. Two independent justifications:*

- *(i) (written proof (not refereed), with the rounding bound by exact computation; new here) the argument below, for $m \in \{6, 7\}$ and every K and t;*
- *(ii) (exact computation, one program sharing* `gen.py`*'s literal translation) only for the two certified instances,* `C6UPPER/encb/compare_ab2.py` *and* `C7UPPER/tests/compare_ab.py` *compared, for every triple orbit having a clause in either encoder and every pattern of its layers, the set of orientation literals forced by A's angular and nesting clauses with that of B's clauses without chain variables: at* $(6, 8, 11)$ 10,417 *clauses each,* 5,297 *orbits,* 23,852 *orbit–pattern pairs,* 0 *mismatches (*`C6UPPER/logs/encb_compare.log`*); at* $(7, 8, 10)$ 74,099 *clauses each,* 6,221 *orbits,* 185,549 *pairs,* 0 *mismatches (*`C7UPPER/logs/repro_validation.log`*). With Lemmas 1.18 and 1.20, an A clause whose premises are true under α_S forces a literal that a B clause with true premises also forces, hence a true one; an A clause with a false premise is satisfied.*

Proof of (i). `fam_nest` emits the same nesting units as encoder B (Lemma 1.20). In `fam_angle`, a pattern is a permutation `perm` of M_c with 0 first, plus H bits for odd m; its premises are the literals $[\psi_{\mathrm{perm}(r)} < \psi_{\mathrm{perm}(r+1)}]$ for consecutive entries not involving layer 0 (`ylit` makes those constant) and the H literals. Since $\psi_0 = 0$ is the least phase, all premises are true under α_S iff `perm` is the ψ-order of M_c and the bits are the h_L; the emission rule is the same as in Lemma 1.18. For the pattern of S, `patterns` supplies phases $\psi_L = u\, q_L/(n+1)$, $n = |M_c|$, $q_L \in \{0, \dots, n\}$ increasing along `perm`, and $\theta_L = \psi_L + (1 - [\mathrm{bit}])\pi/m$ for odd m. These realise the same ψ-order and the same h as S on M_c. Lemma 1.15 applies to both phase assignments on M_c (the sampled ψ_L are distinct and $\theta_L \in [0, 2\pi/m)$), and they give the six objects of c the same keys. So the objects have the same cyclic word, the same counterclockwise order of points and the same block as in S. `angle_config` sorts the three directions, forms the gaps, returns `degenerate` if a gap is within 10^{-9} of 0 or π, `in` with the sign of the sorting permutation if all gaps are $< \pi$, and otherwise the points in order starting after the largest gap; by Lemma 1.16 these are exactly B's three cases and outputs, *provided* the floating-point decisions agree with the exact ones. They do: every exact direction is an integer multiple of $\delta_0 = u/(n+1)$, and so is π ($\pi = m(n+1)\delta_0$ for odd m, $\frac{m}{2}(n+1)\delta_0$ for even m); every exact direction is at most $2\pi - \delta_0$; and the three points have distinct directions (distinct ψ for distinct layers, distinct k in one layer). So each exact gap equals π or lies at distance at least $\delta_0 \ge \pi/(4m)$ from both 0 and π (exactly, the least such distance is $\pi/12$ for $(m, t) = (6, 11)$ and $\pi/28$ for $(7, 10)$). The float inputs depend only on m, on $n \in \{1, 2, 3\}$, on q_L, on the bit and on k, not on K or t. Re-evaluating `gen.py`'s float expressions in all 54 ($m = 6$) and 105 ($m = 7$) such cases and comparing with the exact angles (π enclosed by 36-digit rationals) shows two things (exact computation, `k8_soundness_checks.py`). First, each value `theta[i] + 2*math.pi*k/m` that `angle_config` reduces is a float in $[0, 2*\mathrm{math.pi})$, so the reduction `% (2*math.pi)` returns it unchanged (a wrap-around would have produced an error near 2π). Second, the value used is within $8.7 \cdot 10^{-16}$ of the exact direction. Hence the sorted order is the exact one, each computed gap is within 10^{-14} of the exact gap (two angle errors, the error of `2*math.pi`, and at most two roundings of numbers below 8, each below $9 \cdot 10^{-16}$), gaps equal to π (only same-layer antipodal pairs, even m) are reported as

`degenerate`, and all other comparisons with `eps`$= 10^{-9}$ and with π are decided correctly. The representative's classification is then that of S, and the emitted literal is true under α_S exactly as in Lemma 1.18. □

1.8. **The omitted antipodal case.**

Lemma 1.26 (diameter; written proof (not refereed); the formula is not in the review)**.** *Let m be even, q and $-q$ points of one layer, and p another point. Then*

$$\operatorname{orient}(q, -q, p) = -2\det(q, p) = -2|q||p|\sin\big(\operatorname{dir}(p) - \operatorname{dir}(q)\big),$$

so $\operatorname{orient}(q, -q, p) > 0$ *if and only if p lies strictly clockwise from q by less than π. For a generic normalised configuration this is decided by the keys of q, $-q$ and p (Lemma 1.15), i.e. by the phase pattern alone.*

Proof. $\operatorname{orient}(q, -q, p) = \det(-q - q,\, p - q) = -2\det(q, p) + 2\det(q, q) = -2\det(q, p)$, and $\det(q, p) = |q||p|\sin(\operatorname{dir} p - \operatorname{dir} q)$; $\operatorname{dir}(p) \neq \operatorname{dir}(\pm q)$ by Lemma 1.9. □

A triple containing a same-layer antipodal pair has the centre on one of its sides, so it is neither a triangle around the centre nor contained in an open half-plane through the centre. For every pattern of such an orbit, encoder B finds coinciding keys (line 113) and encoder A a gap of π (`degenerate`); neither emits an angular clause. For $(m, t) = (6, 11)$ these are 352 of the 7,634 orbits ($11 \cdot 3 \cdot 64$ triples in orbits of size 6; exact computation, `k8_soundness_checks.py`). A nesting triple $\{(i, 0), (i, 1), (j, k)\}$, $j > i$, contains no same-layer antipodal pair (as $m \geq 4$), and rotations preserve this property, so none of these orbits carries a nesting unit either; their orientations are constrained only through the chain family. By Lemma 1.26 valid clauses for them exist and are omitted. Omitting valid clauses is a weakening: it can only enlarge the set of satisfying assignments, and Theorem 1.24 concerns only the emitted clauses. So the omission is not a source of unsoundness, and the certificates prove unsatisfiability without these clauses. More generally the encoders omit every orientation fact not covered by rules A and B (for instance half-plane triples whose middle point is not on the outermost layer, and the two-point-chord facts of `CHORD20261105`); each omission is a weakening. The chord-lemma family of `CHORD20261105` (written proof (not refereed), each sixfold clause checked by exact computation) contains such diameter clauses; it was not part of the `C6UPPER` or `C7UPPER` formulas. The `gen.py` docstring states the angular dichotomy without this third case (finding of `C6C7REVIEW/SOUNDNESS.md`); the emitted formulas are unaffected.

1.9. **The cube cover and the passage to geometry.**

Lemma 1.27 (cube cover; written proof (not refereed))**.** *Let F be a CNF formula and $x_1, \dots, x_n$ variables. If $F \wedge \ell_1 \wedge \cdots \wedge \ell_n$ is unsatisfiable for each of the 2^n choices $\ell_j \in \{x_j, \neg x_j\}$, then F is unsatisfiable.*

Proof. A model β of F satisfies the cube with $\ell_j = x_j$ if $\beta(x_j)$ is true and $\ell_j = \neg x_j$ otherwise. □

No geometric meaning of the x_j is needed, so the 20 sign patterns of $Y(1,3), Y(1,4), Y(2,3), Y(2,4), Y(3,4)$ that contradict transitivity need no separate argument (their cubes were refuted anyway). For the sixfold split, F is the base formula plus the unit $Y(1,2)$, and $x_1, \dots, x_5$ are $Y(1,3), Y(1,4), Y(2,3), Y(2,4), Y(3,4)$: variables $2, 3, 10, 11, 18$ in encoder B and $7{,}636, 7{,}637, 7{,}644, 7{,}645, 7{,}652$ in encoder A (by the variable numbering described in the paragraph after Definition 1.12). That each of the 32 cube files of each encoder is the base formula byte for byte, with adjusted header, plus $Y(1,2)$ and the five literals of its pattern, and that the 32 patterns are distinct, was checked by `C6UPPER/tools/verify_cover.py` and by the assertion-independent `C6C7REVIEW/strict_cover.py` (exact computation). Only the variable of the unit $Y(1,2)$ must have its geometric meaning; it is fixed by the source (Theorem 1.24).

Corollary 1.28 (from unsatisfiability to the Lean propositions; written proof (not refereed), not in the review, with the cited Lean-checked reduction)**.** *Let F_A, F_B be as in Theorem 1.24.*

- *(a) If, for one encoder, the* 32 *formulas "base* $(6, 8, 11)$ *+ $Y(1,2)$ + cube" are all unsatisfiable, then* `Sixfold66` *holds; with* `sixfold_octagon` *(Lean-checked), every sixfold rotationally symmetric set of at least* 61 *points in general position contains a convex octagon.*
- *(b) If $F_A(7, 8, 10)$ plus $H(1)$, or $F_B(7, 8, 10)$, is unsatisfiable, then* `Sevenfold70` *holds; with* `sevenfold_octagon` *(Lean-checked), every sevenfold rotationally symmetric set of at least* 65 *points in general position contains a convex octagon.*

Proof. (a) Suppose `Sixfold66` fails: some T in general position, invariant under the rotation by $2\pi/6$ about a point $c \notin T$, with $|T| = 66$, has no convex octagon. Translating c to O preserves these hypotheses (a translation is an affine bijection; Lemma 1.4), so $T - c$ is $(6, 8, 11)$-admissible. By Theorem 1.24(a), resp. (b), the base formula of the encoder plus $Y(1,2)$ is satisfiable; by Lemma 1.27, so is one of the 32 cube formulas, a contradiction. (b)

is the same with 70 points, $(7, 8, 10)$-admissibility and $H(1)$; for F_B the unit may be dropped, since a model of F_B plus $H(1)$ is a model of F_B. □

The hypotheses of the corollary are supplied by certificates (cake_lpr, `s VERIFIED UNSAT`). Their status, including how many verdicts are bound to file hashes by a replay against freshly generated formulas, is stated in the companion section and is not repeated here, since it changes as the replay proceeds. The link from the certified files to the programs analysed here is that fresh runs reproduce the formula files byte for byte (hashes above; `C6UPPER/logs/regen.log`, `C6C7REVIEW/REPLAY.md`).

Ingredient	Status
Strict nesting, radius order, centre exclusion (Lemmas 1.6, 1.7); reduction to `Sixfold66`, `Sevenfold70`	Lean-checked (`LAYERLEAN`)
Layer lemma (Lemma 1.2)	Lean-checked core; the assembly of the peeled layers is a written proof (not refereed), not in the review
Signs of rules A and B (Lemma 1.16(b),(c)); chain soundness (Lemma 1.21, geometry)	Lean-checked
Collinearity with the centre (Lemma 1.9)	written proof (not refereed), new here
Convex position via orientations; affine images (Lemmas 1.10, 1.4)	written proof (not refereed), standard; not in the review
Genericity and perturbation (Lemma 1.11)	written proof (not refereed); proves in full the review's hyperplane statement
Orbit variables and stabilisers; key order; circular dichotomy (a); premises (Lemmas 1.14, 1.15, 1.16, 1.17)	written proof (not refereed)
Code-to-clause correspondence for both encoders (Lemmas 1.18–1.21, Theorem 1.24)	written proof (not refereed) by reading the source
Reflection unit; cube cover (Lemmas 1.23, 1.27)	written proof (not refereed)
Encoder A's float classification, $m \in \{6, 7\}$ only (Proposition 1.25)	exact computation (comparison with B at the two instances; rounding bound), and written proof (not refereed), new here
Cube files equal base plus units; orbit and pattern counts; formula hashes	exact computation
Omitted antipodal clauses (Lemma 1.26)	written proof (not refereed), formula not in the review; a weakening only
Passage to `Sixfold66`, `Sevenfold70` (Corollary 1.28)	written proof (not refereed), not in the review
Unsatisfiability of the formulas	certificate (companion section)
Not needed for the upper bounds: nested m-gons are admissible (Remark 1.5)	written proof (not refereed), new here

TABLE 1. What the implication "certificates ⇒ `Sixfold66`, `Sevenfold70`" rests on.

1.10. **Trust base and gaps.**
Gaps that remain.

(G1) The correspondence between the Python sources and the clause families is established by reading the code at the stated hashes; there is no formal semantics of the programs, and the Python interpreter is trusted to run them as written.

(G2) No step from a DIMACS file to `Sixfold66` or `Sevenfold70` is formalized. The results of this section other than the cited Lean theorems are written proofs only.

(G3) Encoder A's angular part rests on binary64 arithmetic. For the two certified instances Proposition 1.25 covers it twice; for other (K, t) with $m \in \{6, 7\}$ only argument (i) applies, and for other m no claim about encoder A is made. Argument (i) is new here and unreviewed, and comparison (ii) is a single program that uses `gen.py`'s own literal translation.

(G4) The written proofs marked *new here* (Lemma 1.9, Remark 1.5 and Proposition 1.25(i)) and those marked *not in the review* (Lemmas 1.10 and 1.4, the assembly in Lemma 1.2, the formula of Lemma 1.26, and Corollary 1.28) have had no internal review. The remaining written proofs expand arguments reviewed on 4 November.

Where the sources only sketched. The docstring of `gen.py` (step 0) calls K-freeness a strict inequality in the parameters; Lemma 1.10 supplies the needed fact that convex position is decided by orientation signs. The review states that the bad phase equalities form a finite union of proper affine hyperplanes near the configuration; Lemma 1.11(b) proves this in full and identifies the perturbed set as an admissible set through Lemma 1.2. The

angular dichotomy is completed by the antipodal case (Lemma 1.15 and Section 1.8), as the review asked. The chain argument of the docstring and of the review (a simple, locally convex fan polygon) is replaced by the Lean-checked proof through positive triples. Stating everything for admissible sets with their peeled layers lets the Lean theorems apply to the layers directly; the link to nested m-gons given abstractly is Remark 1.5, which the upper bounds do not need.

References

[1] Supplementary material: the directories `C7UPPER`, `C6UPPER`, `LAYERLEAN`, `C6C7REVIEW`, `CHORD20261105`, `STATUS20261104` and `WRITEUP/note` under `cloud/es7-2026-10-19/results` of the project repository.